

GITHUB REPOSITORY AND VERSION CONTROL

Implementation of Collaborative Git/GitHub Workflow

Aim

To create a GitHub repository for a team project and demonstrate version control using branches, pull requests, code review, conflict resolution, and README documentation.

Tasks Performed

1. Set up a repository for the team project.
2. Created branches for individual modules and feature development.
3. Committed changes to the appropriate branches.
4. Created and used pull requests to merge changes after review.
5. Resolved merge conflicts during branch integration.
6. Documented the development workflow using the README file.

Implementation Summary

The project was managed using Git and GitHub. Separate branches were used for development work so that changes could be made independently without directly affecting the main branch. Pull requests were used to review and merge the work. A merge conflict was encountered and resolved before the branch was integrated into the main branch. The final GitHub activity demonstrates the collaborative version-control workflow.

Evidence – GitHub Pull Request

The screenshot displays a GitHub Pull Request (PR) for the repository 'octocat / Spoon-Knife'. The PR is titled '1 #41071' and is currently 'Open'. It shows that user '192511064simats-0607' wants to merge 3 commits from the '192511064simats-0607:main' branch into the 'octocat:main' branch. The interface includes tabs for 'Conversation' (1), 'Commits' (3), 'Checks' (1), and 'Files changed' (1). The commit history shows three commits by '192511064simats-0607' from 2 weeks ago: 'Add EX15 collaboration section' (commit 85af6d9), 'Add EX15 section from main branch' (commit d8d13b1), and 'Resolve merge conflict: EX15 Collaboration section (0c9d333)' (commit 0c9d333). A green checkmark indicates 'No conflicts with base branch' and 'Changes can be cleanly merged.' A 'Merge pull request' button is visible. On the right, there are sections for 'Reviewers' (None yet), 'Assignees' (No one—assign yourself), 'Labels' (None yet), 'Projects' (None yet), 'Milestone' (No milestone), and 'Development' (Successfully merging this pull request may close these issues. None yet).

Figure 1: GitHub pull request showing the branch merge, commit history, conflict-resolution commit, and successful merge status.

Result

The GitHub repository version-control workflow was successfully demonstrated. Branches and pull requests were used for collaboration, conflicts were resolved, and the changes were prepared for integration into the main branch.

Repository / Submission

GitHub repository: octocat/Spoon-Knife (repository shown in the submitted GitHub evidence).

Pull Request: #41071 (shown in the submitted screenshot).